

Series and Convergence

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1 Series as limits of partial sums

Given a real sequence $(a_n)_{n \geq 0}$, the symbol $\sum_{n=0}^{\infty} a_n$ denotes the formal infinite sum. Its value, when defined, is the limit of the sequence of partial sums.

Definition 1 (Series and partial sums). The n -th partial sum of (a_n) is

$$s_n = \sum_{k=0}^n a_k.$$

The series $\sum_{n=0}^{\infty} a_n$ is the symbolic object associated with (a_n) ; we say it *converges to* $S \in \mathbb{R}$ iff $s_n \rightarrow S$ as $n \rightarrow \infty$, and write $\sum_{n=0}^{\infty} a_n = S$. Otherwise the series *diverges*.

Definition 2 (Absolute and conditional convergence). The series $\sum a_n$ *converges absolutely* iff the series of non-negative terms $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, the convergence is *conditional*.

It is a standard fact that absolute convergence implies convergence: the partial sums of $\sum a_n$ form a Cauchy sequence, dominated by the Cauchy tails of $\sum |a_n|$. The converse fails, as witnessed by $\sum (-1)^{n+1}/n = \ln 2$ paired with the divergent harmonic series.

Example 3 (Geometric series). For $r \in \mathbb{R}$ with $|r| < 1$, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$. Indeed $s_n = (1 - r^{n+1})/(1 - r) \rightarrow 1/(1 - r)$ as $r^{n+1} \rightarrow 0$. This is the workhorse comparator for the ratio test below.

2 The ratio test

The ratio test reduces convergence of $\sum a_n$ to comparison with a geometric series.

Theorem 4 (Ratio test). Let (a_n) be a sequence of nonzero real numbers and suppose

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists in $[0, +\infty]$. Then:

1. if $L < 1$, the series $\sum_{n=0}^{\infty} a_n$ converges absolutely;
2. if $L > 1$ (including $L = +\infty$), the series diverges;
3. if $L = 1$, the test is inconclusive.

Proof. We treat the three cases.

Case $L < 1$. Choose r with $L < r < 1$. Since $|a_{n+1}/a_n| \rightarrow L < r$, there exists $N \in \mathbb{N}$ such that

$$\left| \frac{a_{n+1}}{a_n} \right| < r \quad \text{for all } n \geq N.$$

Multiplying these inequalities for $n = N, N+1, \dots, N+k-1$ gives

$$|a_{N+k}| < r^k |a_N| \quad \text{for every } k \geq 0.$$

Therefore, for all $n \geq N$,

$$\sum_{m=N}^n |a_m| \leq |a_N| \sum_{k=0}^{n-N} r^k \leq \frac{|a_N|}{1-r}.$$

The partial sums of the non-negative series $\sum |a_m|$ are bounded above, hence converge (a monotone bounded sequence has a limit). Adding back the finitely many terms for $m < N$ shows $\sum |a_n| < \infty$, so $\sum a_n$ converges absolutely.

Case $L > 1$. Choose r with $1 < r < L$ (or $r > 1$ arbitrary if $L = +\infty$). There exists N such that $|a_{n+1}/a_n| > r$ for all $n \geq N$, hence $|a_{N+k}| > r^k |a_N|$. Since $r > 1$, $|a_n| \rightarrow \infty$, so $a_n \not\rightarrow 0$ and the series diverges by the term test.

Case $L = 1$. The series $\sum 1/n$ has $L = 1$ and diverges, while $\sum 1/n^2$ has $L = 1$ and converges. No conclusion can be drawn from L alone. $\square$

Example 5 (Exponential series). For every $x \in \mathbb{R}$, setting $a_n = x^n/n!$ gives

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|}{n+1} \rightarrow 0.$$

Thus $L = 0 < 1$, and Theorem ?? yields absolute convergence of $\sum_{n=0}^{\infty} x^n/n! = e^x$ for every real x . The same argument works for $x \in \mathbb{C}$ and, with $|x|$ replaced by $\|A\|$, for any square matrix A .

3 Power series

A *power series* centred at x_0 is a series of the form $\sum_{n=0}^{\infty} c_n(x - x_0)^n$. Its set of convergence is an interval (possibly a single point or all of $\mathbb{R}$) of *radius of convergence*

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}} \quad (\text{Cauchy--Hadamard}).$$

Inside $|x - x_0| < R$ the series converges absolutely; outside it diverges; behaviour at the endpoints requires a separate test.

This perspective is the bridge from analysis to ML: every smooth activation function, the matrix exponential, and the leading-order expansion of the KL divergence are all instances of power-series approximation.